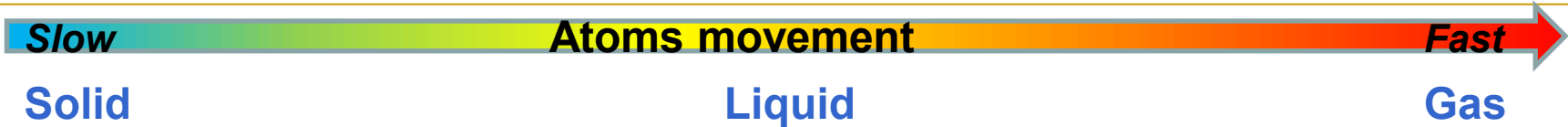


TOPIC 3. DIFFUSION IN SOLIDS

- **Concept of diffusion**
- **Diffusion mechanisms**
- **Steady state diffusion. Fick's 1st law**
- **Non-steady state diffusion. Fick's 2nd law**
- **Factors affecting diffusion**
- **Applications**
- **Problems**

What is Diffusion ?

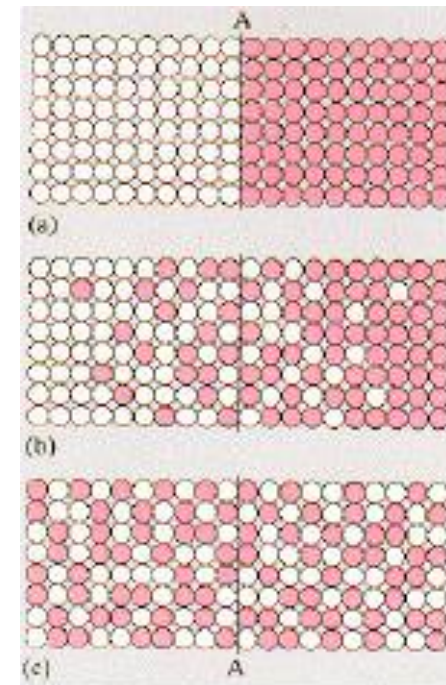
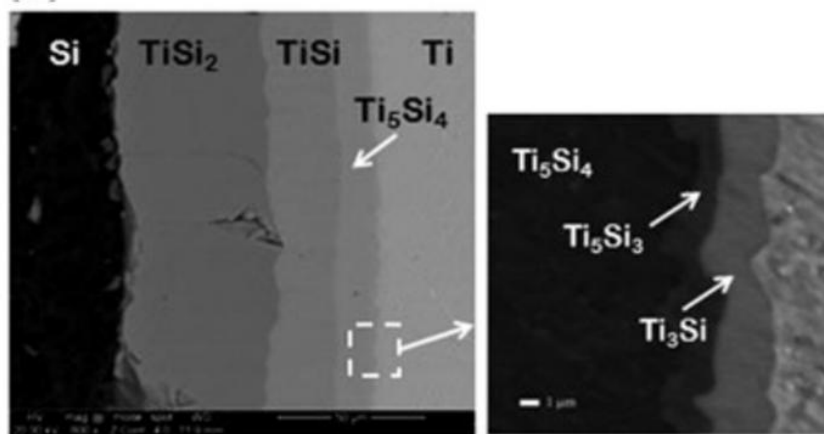
Diffusion (transport phenomena): mechanisms by which material is transported through the material.



Diffusion in solids: movement of atoms within a *solid*

- The thermal vibrations in solids permit *movement of some atoms*
- Phenomena that need **thermal activation**.

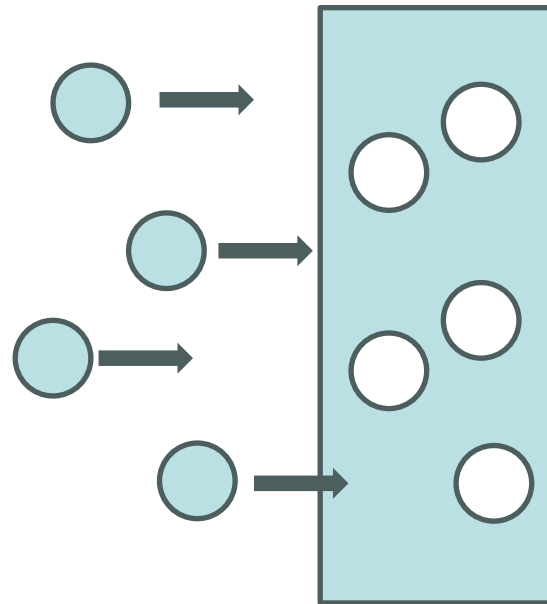
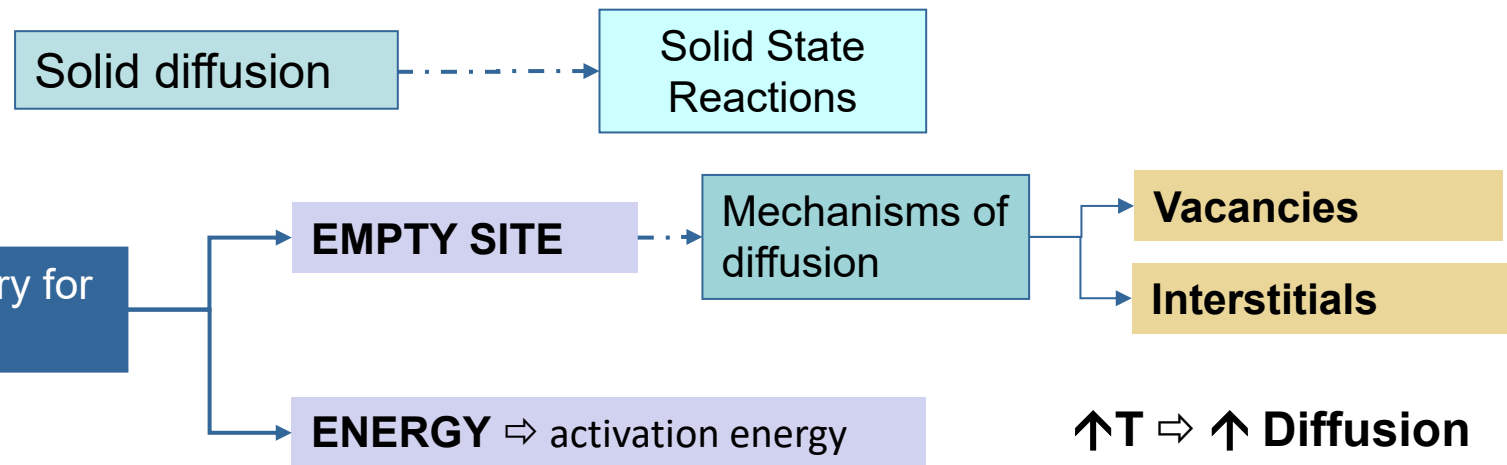
DRIVING FORCE = Composition gradient (equal composition)



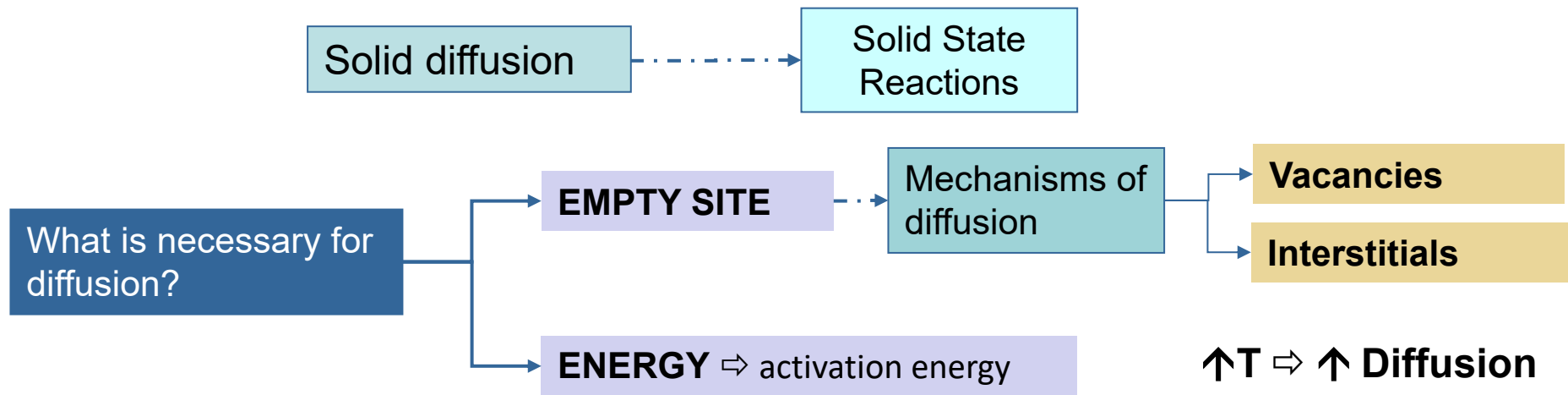
TOPIC 3. DIFFUSION IN SOLIDS

- Concept of diffusion
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DIFFUSION MECHANISMS



DIFFUSION MECHANISMS



- Energy is required to move atoms past one another during diffusion. This energy is the **activation energy** E_a (J/mol or cal/mol).
- Thermal energy should be larger than E_a to produce diffusion.

Diffusivity is governed by an **Arrhenius equation**.

$$D = D_0 e^{-E_a/RT}$$

E_a = Activation energy (J/mol or Cal/mol)
 R = Molar gas constant (J/mol.K or cal/mol.K)
 T = Temperature in Kelvin
 D_0 = constant (Independent of temperature and different for each material)

DIFFUSION MECHANISMS

ACTIVATION ENERGY

Activation energy (E_a) is the **energy required to move** an atom into the new vacancy or interstitial position

- It depends both on the type of atom that diffuses and the material in which the diffusion occurs.

TYPES OF DIFFUSION

1 **Vacancy diffusion** (or substitutional atom diffusion)

1a Self diffusion

1b Interdiffusion

2 **Interstitial diffusion**

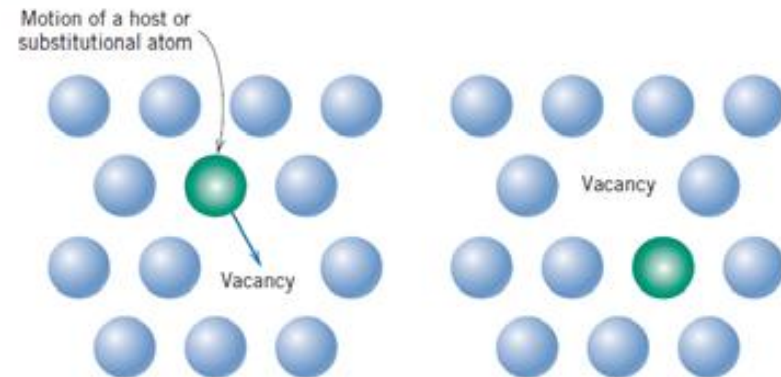
DIFFUSION MECHANISMS

1. VACANCY (OR SUBSTITUTIONAL) DIFFUSION

Movement of atoms into vacant lattice points: displacement of vacancies or defects. Vacancies are defects in equilibrium and are always present

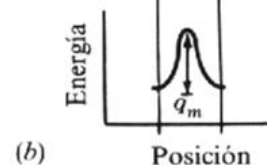
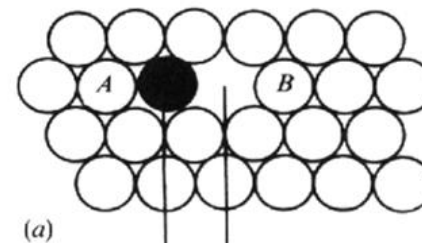
- Its rate strongly depends on the **number of vacancies** in the net, which increases with temperature

$$\text{As } \uparrow T \Rightarrow \left\{ \begin{array}{l} \uparrow [\text{defects}] \\ \uparrow E_{\text{thermal}} < E_{\text{act}} \end{array} \right\} \Rightarrow \uparrow \text{diffusion}$$



1.a Self-diffusion: Exchange of positions of atoms in the metal crystal

Example: If atom 'A' has sufficient activation energy, it moves into the vacancy self-diffusion.



Potential Energy of atoms as a function of their position

DIFFUSION MECHANISMS

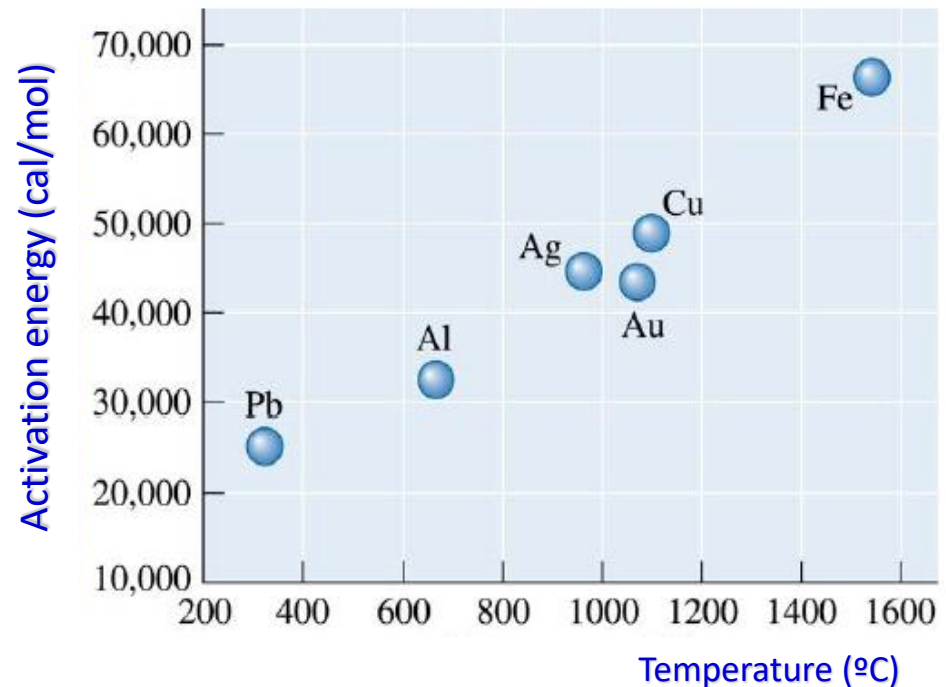
1. VACANCY (OR SUBSTITUTIONAL) DIFFUSION

1.a Self-diffusion: Exchange of positions of atoms in the metal crystal

$$\text{Activation Energy of Self diffusion} = \text{Activation Energy to form a Vacancy} + \text{Activation Energy to move a vacancy}$$

The activation energy for self-diffusion increases as the T_m of the metal increases

- Given that if $T_{\text{melting}} \uparrow \Rightarrow E_{\text{bond}} \uparrow$
- When $\uparrow T_{\text{melting}} \Rightarrow E_a \uparrow$



DIFFUSION MECHANISMS

1. VACANCY (OR SUBSTITUTIONAL) DIFFUSION

1.a Self-diffusion

The activation energy for self-diffusion increases as the T_m of the metal increases

		Melting Temperature (°C)	Crystal structure	Activation energy (kJ/mol)	Activation energy (kcal/mol)
Zinc		419	HCP	91.6	21.9
Aluminium	-	660	FCC	165	39.5
Copper		1083	FCC	196	46.9
Nickel		1452	FCC	293	70.1
α -Iron	+	1530	BCC	240	57.5
Molybdenum		2600	BCC	460	110

DIFFUSION MECHANISMS

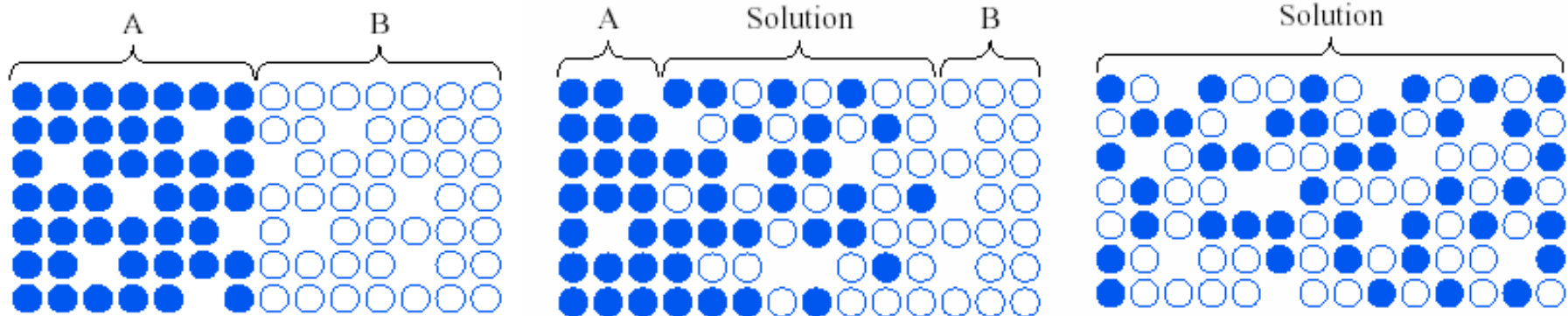
1. VACANCY (OR SUBSTITUTIONAL) DIFFUSION

1.a Interdiffusion : Movement of atoms through a lattice composed of different atoms (as in substitutional solid solutions)

Solid solution rules:
Hume-Rothery

Examples:

- Allows the formation of alloys/intermetallics.
- Important in diffusion welding processes

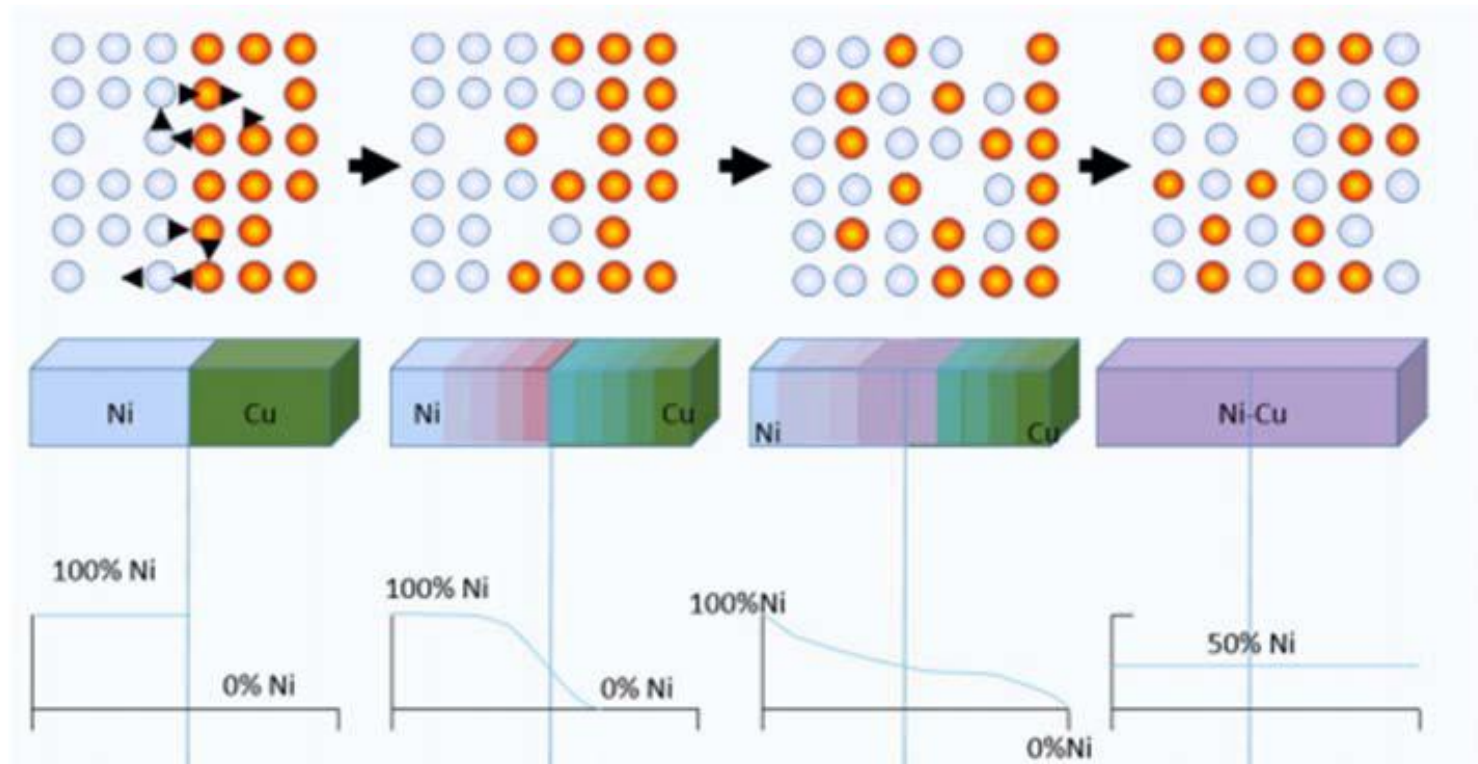


DIFFUSION MECHANISMS

1. VACANCY (OR SUBSTITUTIONAL) DIFFUSION

1.a Interdiffusion

Examples: **Diffusion couple** Copper - Nickel



Before a high-temperature heat treatment

After a high-temperature heat treatment

DIFFUSION MECHANISMS

1. VACANCY (OR SUBSTITUTIONAL) DIFFUSION

1.a Interdiffusion

Examples: **Diffusion couple** Copper - Nickel

Hume-Rothery solution rules for Cu and Ni:

1. Crystal structure:

Both Cu and Ni are **FCC** (face-centered cubic) ✓

2. Atomic radius difference:

$R_{\text{Cu}} = 128 \text{ pm}$

$R_{\text{Ni}} = 124 \text{ pm}$

Difference below 15% ✓

3. Electronegativity:

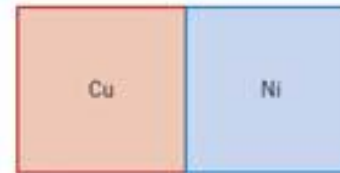
$\text{Cu} \approx 1.90$

$\text{Ni} \approx 1.91$

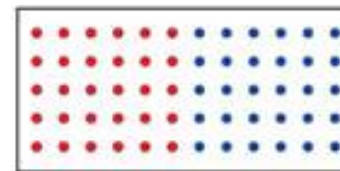
Very similar values ✓

4. Valence:

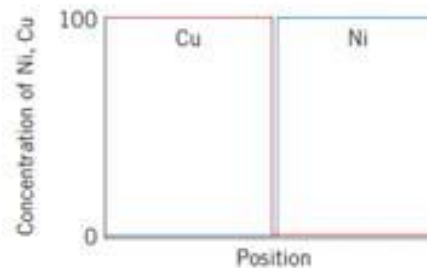
Both have a valence of +2 ✓



(a)

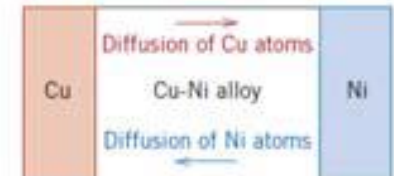


(b)

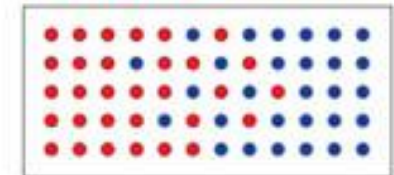


(c)

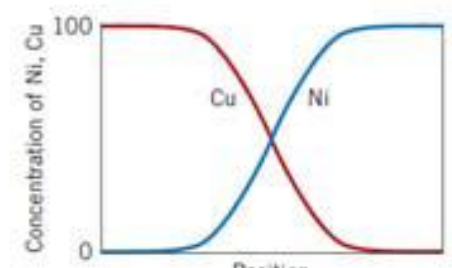
Before a high temperature heat treatment



(a)



(b)

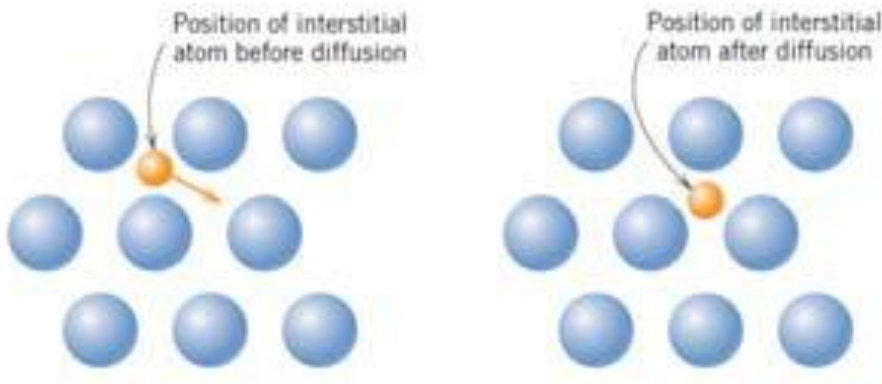


(c)

After a high temperature heat treatment

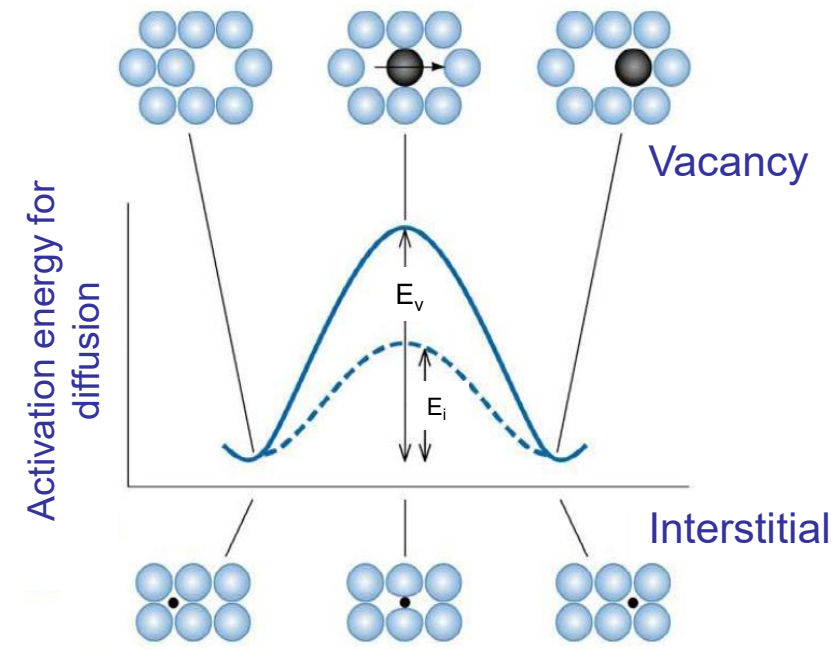
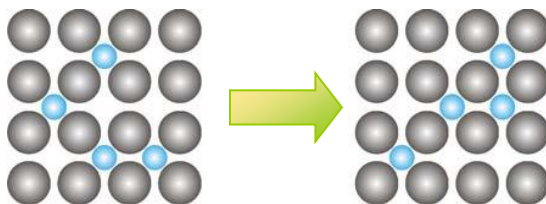
DIFFUSION MECHANISMS

2. INTERSTITIAL DIFFUSION: Atoms in interstitial positions are displaced to other interstitial positions **without displacing** permanently any atoms of the matrix



The atoms that diffuse interstitially are small with respect to the matrix (i.e.: in metals: C, H, O, N)

- ✓ This type of diffusion **needs less energy than vacancy diffusion**
- ✓ It is **faster**
- ✓ There are **more interstitial sites than vacancies**





TOPIC 3. DIFFUSION IN SOLIDS

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- Problems

FICK'S DIFFUSION LAWS

HOW DO WE QUANTIFY THE AMOUNT OR RATE OF DIFFUSION?

The rate at which the atoms diffuse in a material can be measured by a diffusion **flux**, **J** defined as the n° of atoms that passing through a plane of unit area per unit time

Flux density

$$J = Flux = \frac{\text{moles (or mass) diffusing}}{(\text{area})(\text{time})} \left(\frac{\text{mol}}{\text{cm}^2\text{s}} \right) \text{ or } \left(\frac{\text{kg}}{\text{m}^2\text{s}} \right)$$



➤ FICK'S DIFFUSION LAWS

Without the L!!!

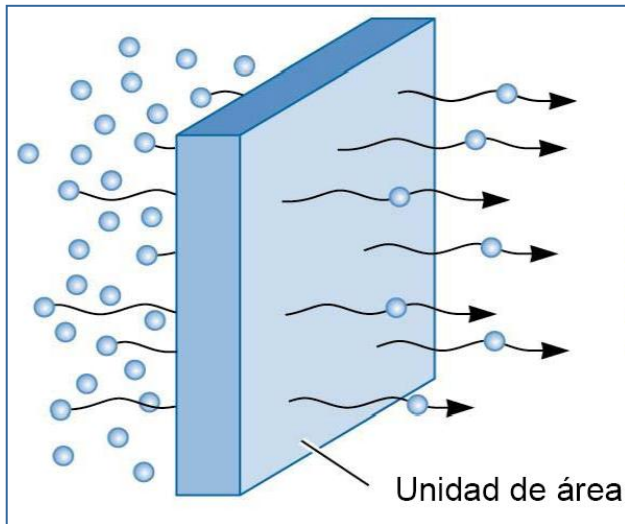
Matter flux {

- Steady-state** conditions → J unchanged over t (Fick's 1st Law)
- Non steady-state** conditions → J changed over t (Fick's 2nd Law)

STEADY STATE DIFFUSION : Fick's 1st law

Steady state conditions: The concentration of solute atoms at any point does not change with time. The diffusion flux does not change with time

$$\frac{\partial J}{\partial t} = 0$$



$$J = -D \frac{\Delta C}{\Delta x}$$

It is created putting in contact 2 materials with different composition.

**DRIVING FORCE =
Composition gradient**

J = nº atoms (or mass) per unit surface and time
along distance x (at/m²s or kg/m²s)

$\Delta C / \Delta x$ = concentration gradient in x direction (at/m³
or kg/m³)

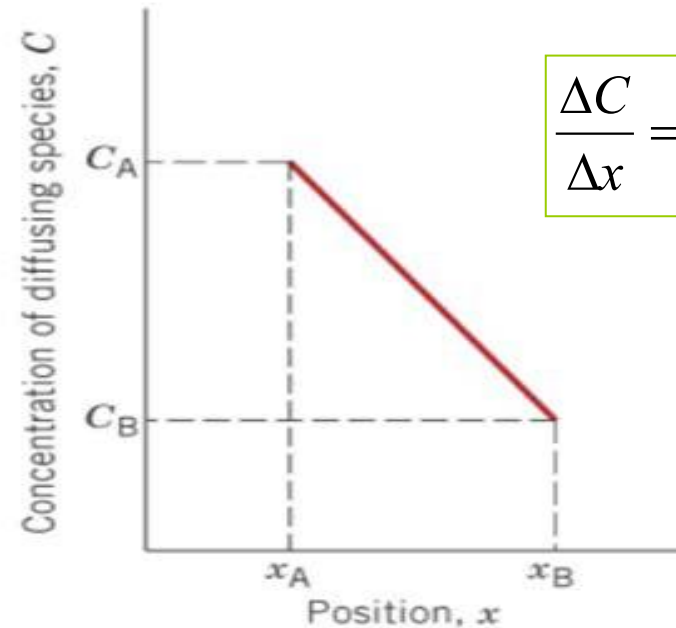
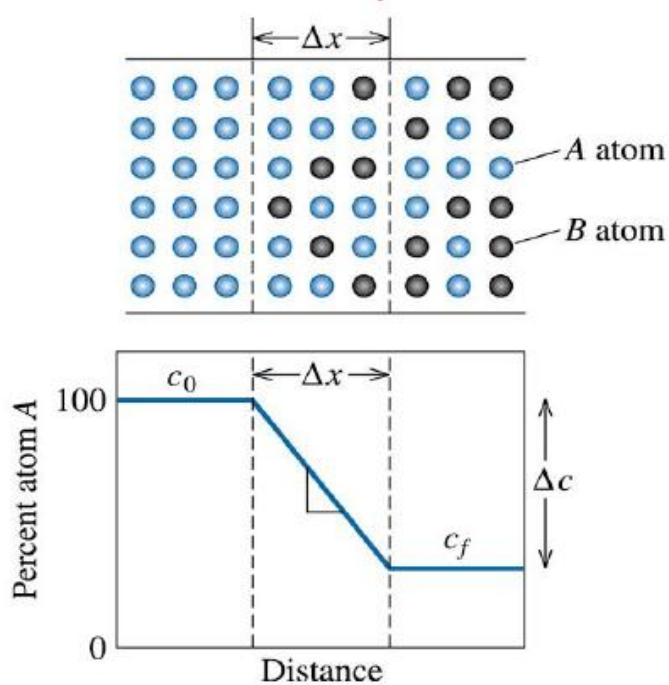
D = **diffusivity or diffusion coefficient** (m²/s)

$$\frac{\Delta C}{\Delta x} = \text{concentration gradient}$$

STEADY STATE DIFFUSION : Fick's 1st law

Steady state conditions: The concentration of solute atoms at any point does not change with time. The diffusion flux does not change with time

$$\frac{\partial J}{\partial t} = 0$$



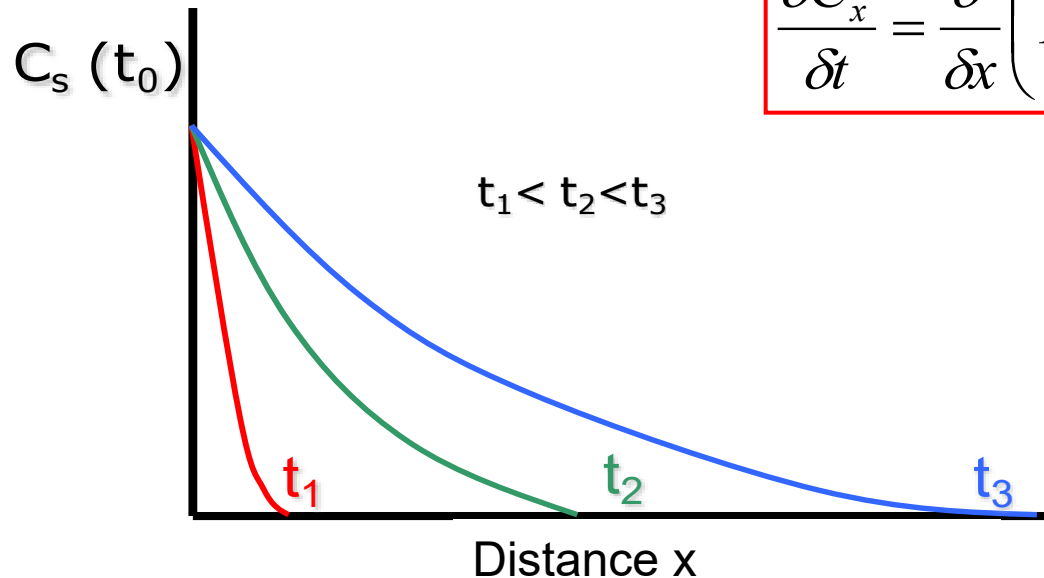
$$\frac{\Delta C}{\Delta x} = \frac{C_{ext} - C_{int}}{x_1 - x_2}$$

J at a T is constant only if the compositions are constant on each side →
if the composition gradient is constant

NON-STEADY STATE DIFFUSION : Fick's 2nd law

For the majority of materials $\Rightarrow$ diffusion is **non-steady state (transient)**: The concentration of solute atoms at any point of the material changes with time.

Non stationary state $\Rightarrow$ The substances come to an end through extinction

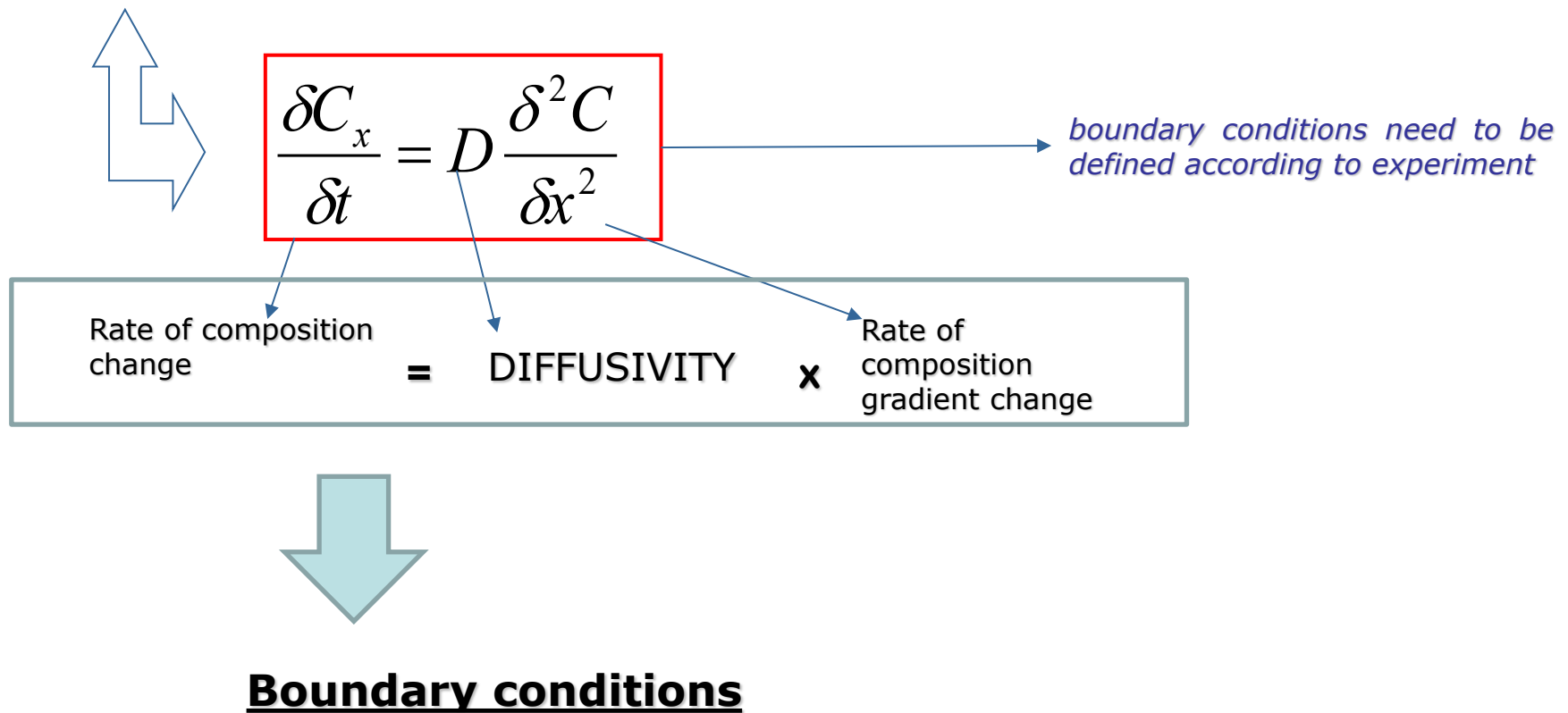


$$\frac{\partial C_x}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

$C_x \neq \text{constant}$
 $D \neq f(t)$

NON-STEADY STATE DIFFUSION : Fick's 2nd law

HYPOTHESIS: if $D \neq f(C)$ i.e. diffusion coeff. is independent of composition (there is no interaction between diffusing atoms)



NON-STEADY STATE DIFFUSION : Fick's 2nd law

Example: diffusion of a gas (with constant surface concentration) in the interior of a semi infinite solid (carbon diffusion in steel, saturation of a metal with atmospheric gases)

For Boundary conditions:

semi-infinite solid

x = distance from surface into solid

$t=0$, $C=C_0$ at $0 \leq x \leq \infty$.

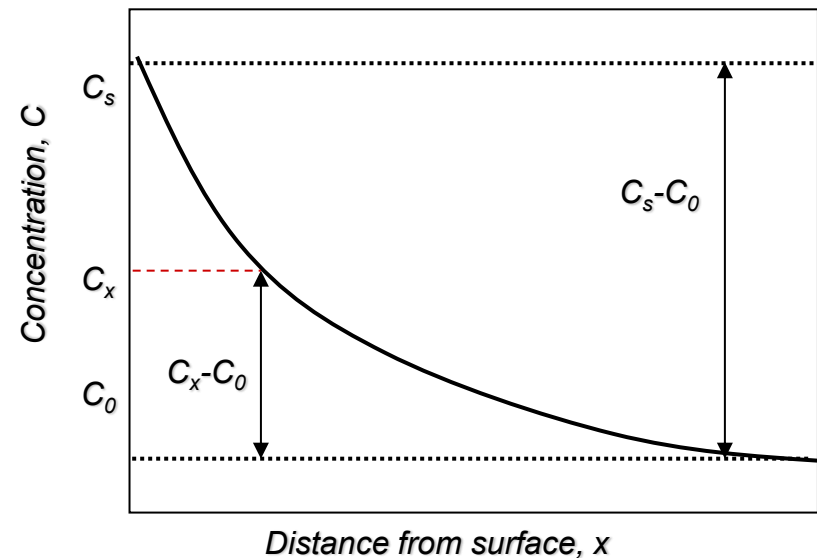
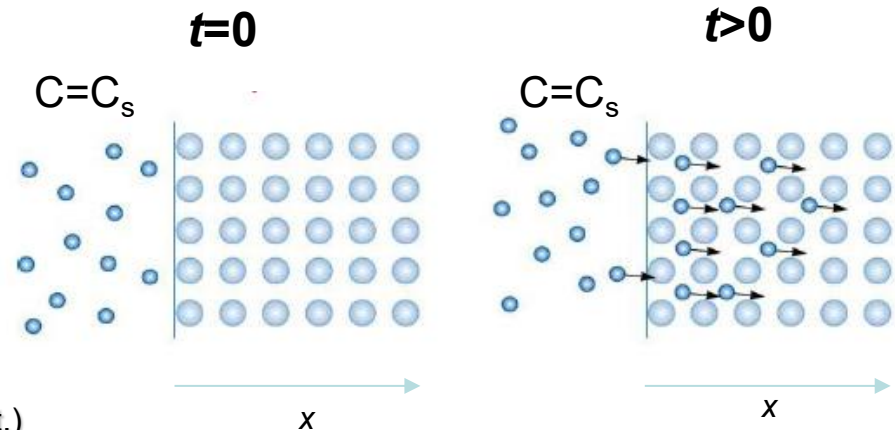
$t>0$, $\rightarrow C=C_s$ at $x=0$ (surface concentration constant.)

$\rightarrow C=C_0$ at $x=\infty$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

C_x = concentration at a distance x after time t

Erf = Gaussian error function



NON-STEADY STATE DIFFUSION : Fick's 2nd law

Example: diffusion of a gas (with constant surface concentration) in the interior of a semi infinite solid (carbon diffusion in steel, saturation of a metal with atmospheric gases)

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$\rightarrow C=C_0$ at $x=\infty$

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

C_x = concentration at a distance x after time t

Erf = Gaussian error function

z	$\operatorname{erf}(z)$	z	$\operatorname{erf}(z)$
0.00	0.0000	0.70	0.6778
0.01	0.0113	0.75	0.7112
0.02	0.0226	0.80	0.7421
0.03	0.0338	0.85	0.7707
0.04	0.0451	0.90	0.7969
0.05	0.0564	0.95	0.8209
0.10	0.1125	1.00	0.8427
0.15	0.1680	1.10	0.8802
0.20	0.2227	1.20	0.9103
0.25	0.2763	1.30	0.9340
0.30	0.3286	1.40	0.9523
0.35	0.3794	1.50	0.9661
0.40	0.4284	1.60	0.9763
0.45	0.4755	1.70	0.9838
0.50	0.5205	1.80	0.9891
0.55	0.5633	1.90	0.9928
0.60	0.6039	2.00	0.9953
0.65	0.6420		

TOPIC 3. DIFFUSION IN SOLIDS

- Concept of diffusion
- Diffusion mechanisms
- Steady state diffusion. Fick's 1st law
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- **Factors affecting diffusion**
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- Problems

FACTORS THAT INFLUENCE DIFFUSIVITY

DIFFUSIVITY

- Depends **on both the atoms that diffuse and the matrix**
- It is **constant for a particular system** (solute+solvent) and **at a specific temperature**

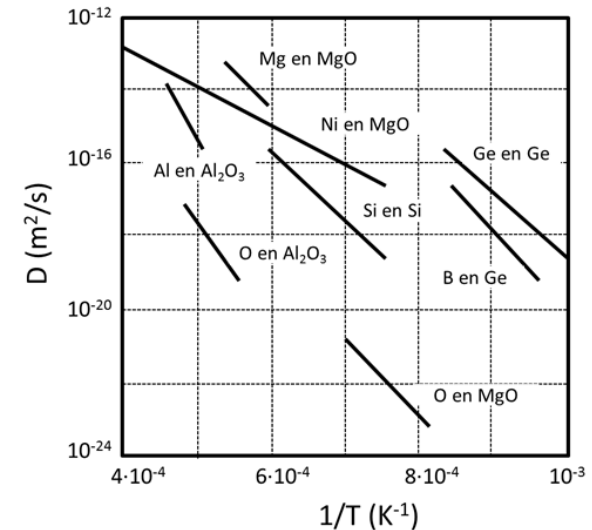
Solute	Solvent Structure solvent	Diffisivity at 500°C m ² /s	Diffisivity at 1000°C m ² /s
Carbon	Iron FCC	5 x10 ^{-15*}	3 x10 ⁻¹¹
Carbon	Iron BCC	10 ⁻¹²	2x10 ^{-9*}
Iron	Iron FCC	2x10 ^{-23*}	2x10 ⁻¹⁶
Iron	Iron BCC	10 ⁻²⁰	3x10 ^{-14*}
Nickel	Iron FCC	10 ⁻²³	2x10 ^{-16*}
Manganese	Iron FCC	3X10 ^{-24*}	10 ⁻¹⁶
Zinc	Copper	4X10 ⁻¹⁸	5x10 ⁻¹³
Copper	Aluminium	4X10 ⁻¹⁴	-
Copper	Copper	10 ⁻¹⁸	2x10 ⁻¹³
Silver	Silver (crystal)	10 ⁻¹⁷	-
Silver	Silver (grain boudnary)	10 ⁻¹¹	-
Carbon	Titanium HCP	3x10 ⁻¹⁶	2x10 ^{-11*}
*metastable phases			

FACTORS THAT INFLUENCE DIFFUSIVITY

a) Diffusion mechanism: Atom size

C diffuses interstitially in Fe- γ ($D_{1000^\circ\text{C}} = 3 \cdot 10^{-11} \text{ m}^2/\text{s}$)

Fe diffuses by vacancy diffusion in Fe- γ ($D_{1000^\circ\text{C}} = 2 \cdot 10^{-16} \text{ m}^2/\text{s}$)

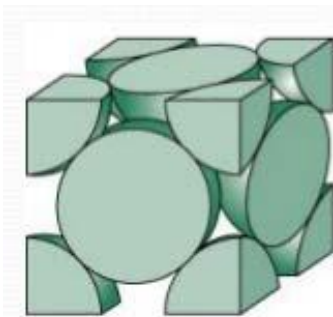


b) Type of crystalline structure of the matrix lattice (solvent)

$$D_{(\text{C in Fe-}\gamma, \text{ FCC})} = 5 \cdot 10^{-15} \text{ m}^2/\text{s} < D_{(\text{C in Fe-}\alpha, \text{ BCC})} = 10^{-12} \text{ m}^2/\text{s}$$

packing factor (BCC) < packing factor (FCC)

$d_{\text{interatomic}} (\text{BCC}) > d_{\text{interatomic}} (\text{FCC})$

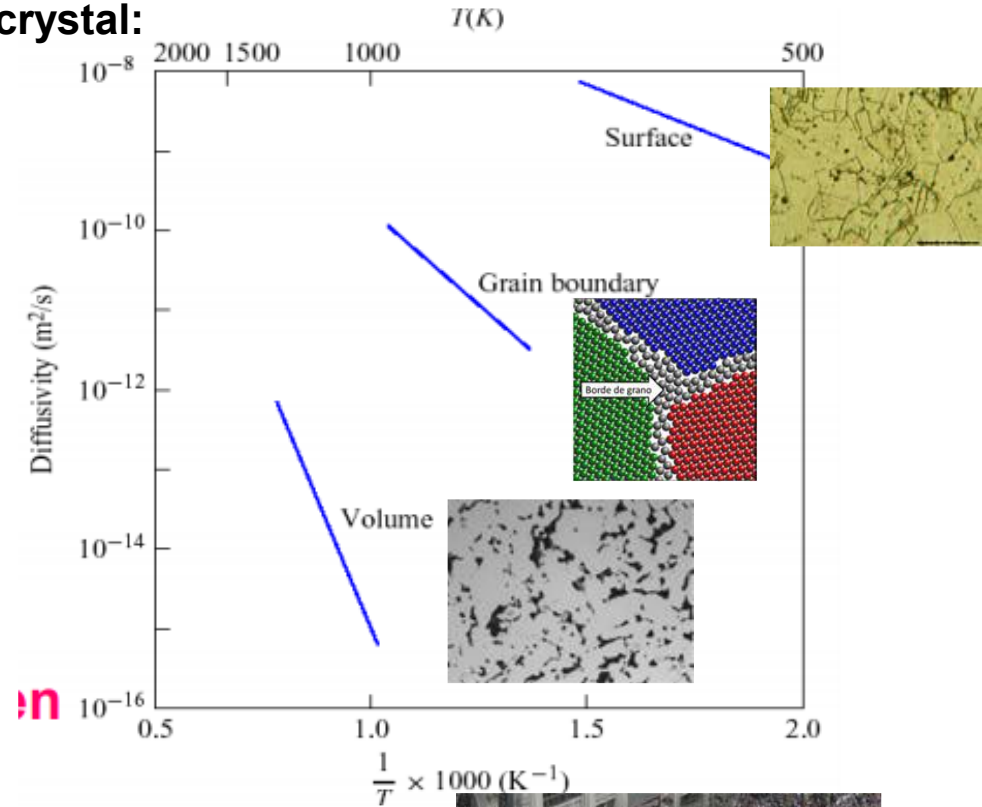


FACTORS THAT INFLUENCE THE DIFFUSIVITY

c) Type of imperfections or defects in the crystal:

Open structures (more holes) $\Rightarrow \uparrow D$

$$D_{\text{surface}} > D_{\text{grain boundary}} > D_{\text{volume}}$$



d) Concentration of species that diffuse

When we $\uparrow$ the concentration of solute atoms we modify D



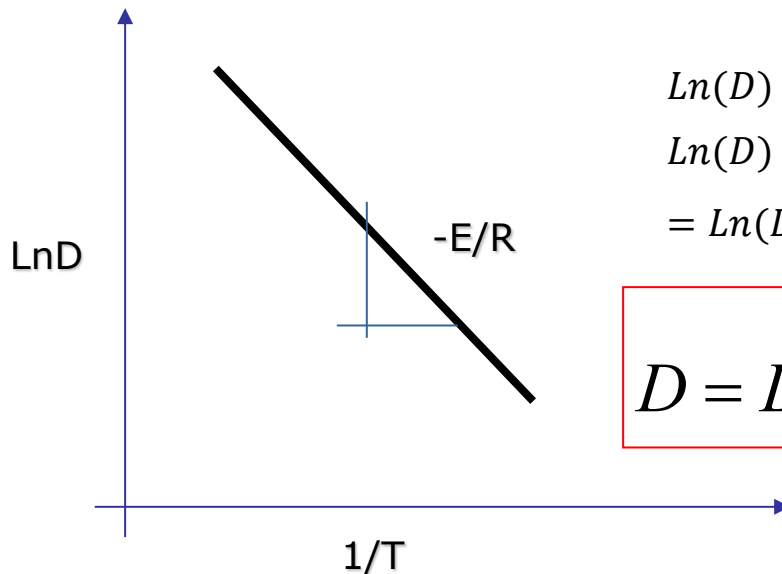
FACTORS THAT INFLUENCE THE DIFFUSIVITY

e) Temperature

$\uparrow T \Rightarrow \uparrow D$

Diffusion of atoms: when $\uparrow T$, $\uparrow$ amplitude of vibration of atoms $\Rightarrow$ greater probability of jump $\Rightarrow D \uparrow$

$D = f(T)$ Experimentally we find there is an Arrhenius type dependence :



$$\ln(D) = y_0 - \frac{E}{R \cdot T}$$

$$\ln(D) = \ln(D_0) - \frac{E_D}{R \cdot T}$$

$$D = D_0 e^{-\frac{E_D}{RT}}$$

D = diffusivity (m^2/s)

D_0 = proportional coeff.
(frequency factor)

E_D = activation energy necessary to
produce the diffusive
movement of 1 mol of atoms

R = gas constant

T = Temperature (K)

DIFFUSION: SUMMARY

- **Diffusion rate:** 'speed' of diffusion of observed species through the material
- Depends on both diffusing species and the characteristics of the material in which the diffusion occurs
- Factors that affect the speed of diffusion:

DIFFUSION is FASTER in...

- Open crystalline structures
- Lower density materials
- Structures with a large number of defects
- Neutral diffusing species
- Materials with low T_m
- Higher temperatures
- Small species

DIFFUSION is SLOWER in...

- Compact crystalline structures
- Higher density materials
- Structures with small number of defects
- Charged diffusing species
- Materials with high T_m
- Low temperature
- Large species



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INDUSTRIAL APPLICATIONS EXAMPLES

- Cementation of gears $\text{CH}_4 - \text{H}_2$
- Sintering
- Transport of gases through polymer films
- Fabrication of integrated circuits with Si wafers: diffusion of impurities.
- Si powder nitriding: Si_3N_4
- Diffusion soldering



CEMENTATION- CASE HARDENING

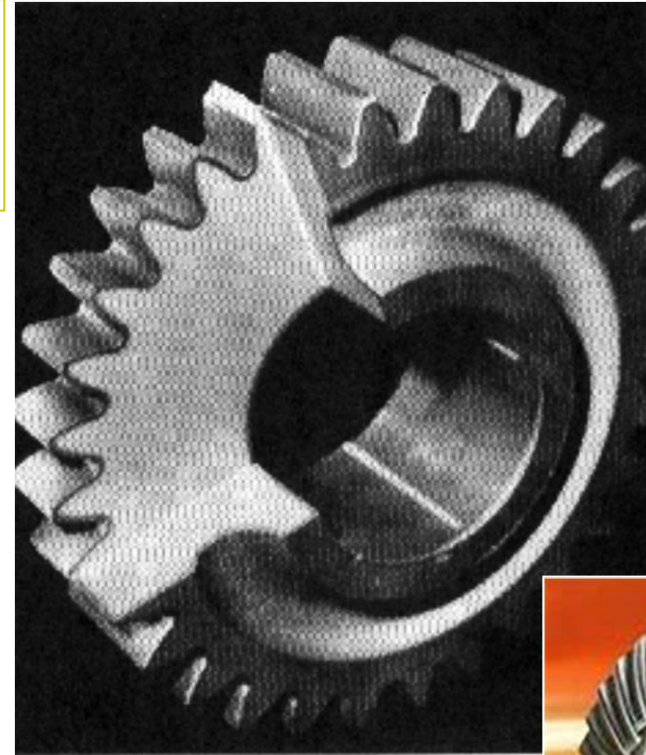
Cementation –Case hardening

(Interstitial diffusion) Diffusion of C atoms in the surface of a steel component .

The C atoms “block” the movement of dislocations, deform the lattice, make more difficult plastic deformation → **Hardening**

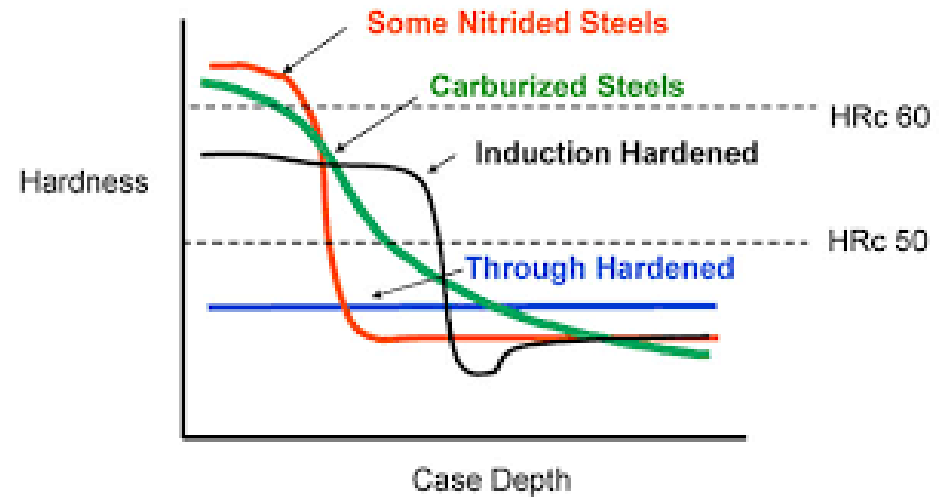
The excess of C in the surface leaves a state of residual stresses in **compression**

- Typical for gears and shafts.
- Inner core must be tough for fracture strength
- Outer core must be hard for wear resistance
- Machining is done in the “soft” state and hardening by a gas carburizing process is done afterwards.



INDUSTRIAL APPLICATIONS EXAMPLES

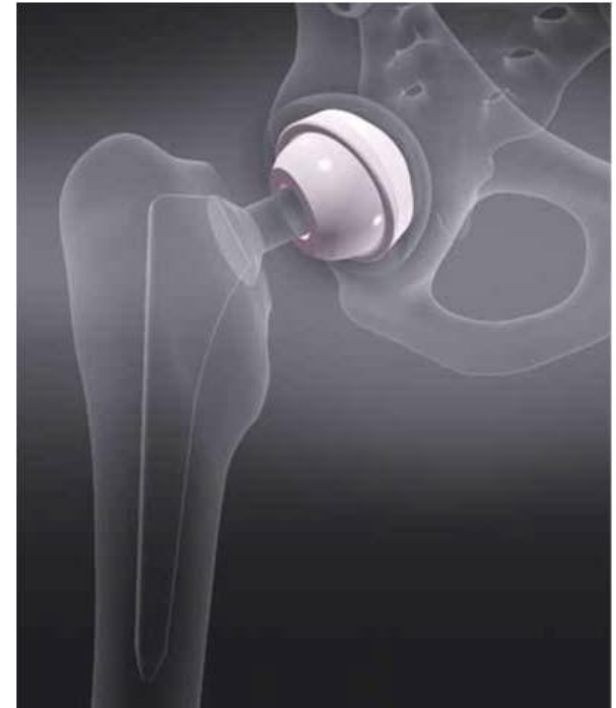
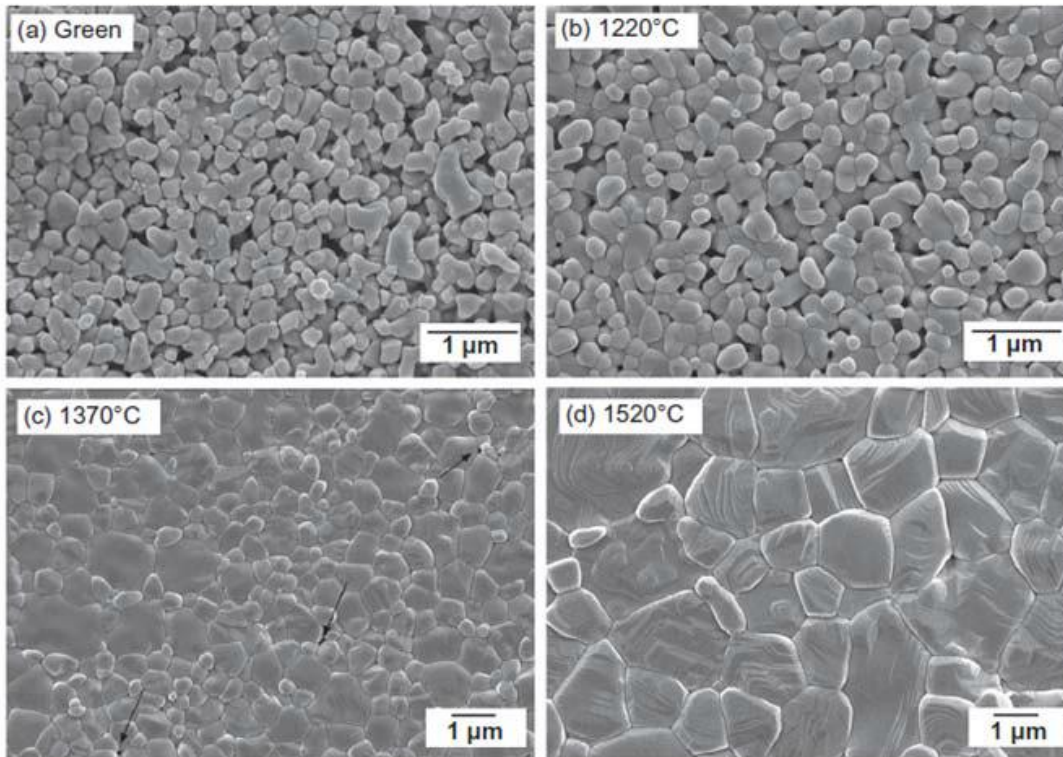
Si powder nitriding: Si_3N_4



INDUSTRIAL APPLICATIONS EXAMPLES

Sintering (e.g. ceramics)

Definition: Mass transport process, thermally activated, conducting to the chemical bonding among particles, and to the reduction of porosity

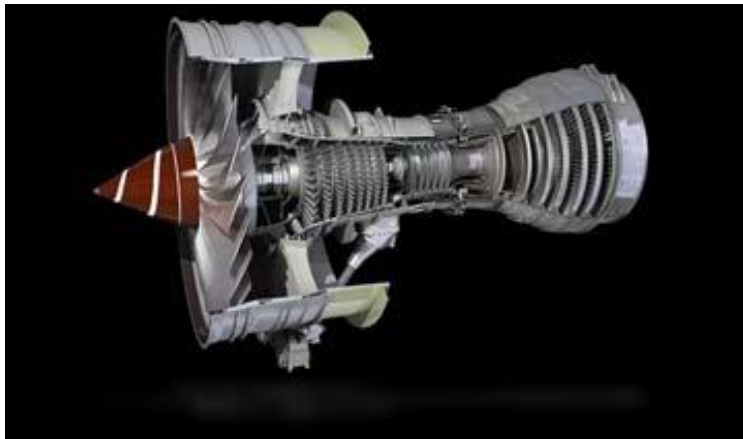


INDUSTRIAL APPLICATIONS EXAMPLES

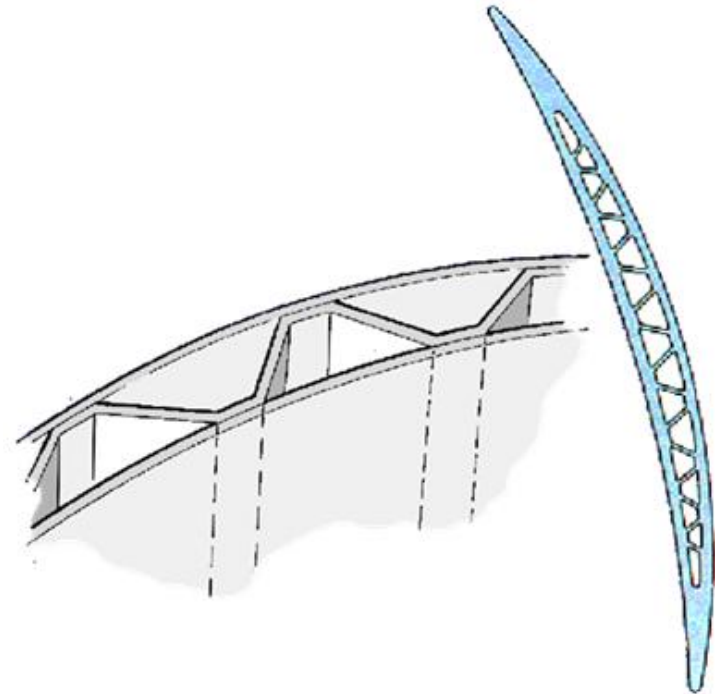
Diffusion bonding

Cowlings and fan blades for aeronautics

Rolls Royce: Trent 500 wide-chord fan protects the engine against foreign object damage



1995



DB/SPF
construction

INDUSTRIAL APPLICATIONS EXAMPLES

Transport of gases through polymer films

Polymers as barrier materials for food packaging

Water vapour:

- Fresh food products dry out. Dried foods clump together, deep fried products such as snacks lose their crispness
- A good barrier has $< 10 \text{ g/m}_2 \cdot \text{day}$, measured at 23°C and 90% RH.

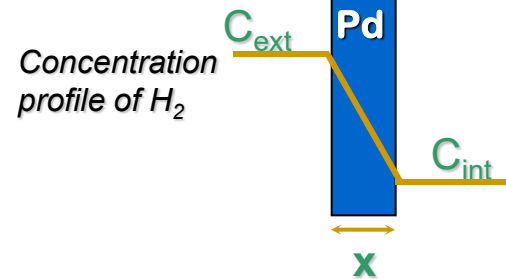
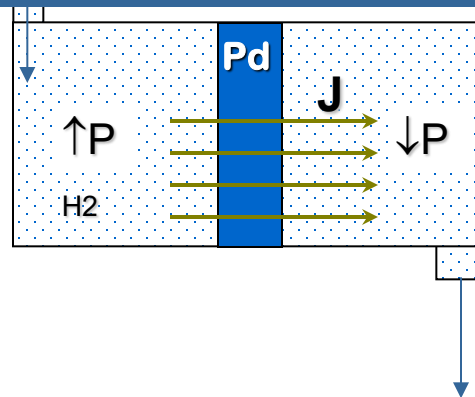


Oxygen and other gases

- Half-baked bakery goods and cheese grow mould. Fresh meat products become discolored. Bacterial growth in fresh products. Sliced meat turn grey under the shop lighting. Products with a high fat content become rancid, especially when exposed to light.
- Best atmospheres depend on food.

STEADY STATE DIFFUSION : example

Example: Diffusion of inert gas through a layer of Pd. H₂ Purification



$$\frac{\Delta C}{\Delta x} = \frac{C_{ext} - C_{int}}{x_1 - x_2}$$

Determine the thickness of a plate of Pd with transversal area of 0.2m² so that it can purify 1.733·10⁻³ kg/h of hydrogen, if the hydrogen concentration at the side of high pressure of the plate is 1.5 kg/m³ and at the side of low pressure is 0.3 kg/m³. The diffusion coefficient of hydrogen in Pd is 1·10⁻⁸ m²/s.